Teaching notes

Students appreciate opportunities to demonstrate their practical skills.

This activity is designed to be carried out over a period of 1 – 2 lessons at the end of a topic on light.

If possible, keep group sizes small, 2-3 students, to ensure that everyone is occupied. To make it manageable, assess each group rather than individual students.

Share with students what you will be looking for and explain that each skill is awarded a mark out of 10 or 20.

There are a total of 100 marks.

30–39 = Level 3; 50–59 = Level 5; 80+ = Level 8

Apparatus

- ray/light boxes
- plane mirrors
- concave and convex mirrors
- a variety of transparent blocks
- prisms
- concave and convex lenses
- plain paper for diagrams

If the amount of equipment available is an issue then each group's sheet could be marked so that they each start with a different practical.

Please ensure you carry out your own risk assessment.

Skills development

- planning
- manipulation of equipment
- testing ideas
- developing problem solving strategies
- team work and communication

Student sheet – introduction

Your group is going to carry out a series of practical activities. Your group will be assessed on the following criteria.

Success criteria

1.	Planning	➢ what do you need to carry out the activity?	**(10)**
2.	Choosing the correct equipment		**(10)**
3.	Team work	➢ make sure everyone has a role.	**(10)**
4.	Setting up	➢ is your equipment in the right position?	**(10)**
5.	Ray diagrams	➢ remember the rules!	**(20)**
6.	Describing	➢ can you tell others what is happening? ➢ can you explain how to work safely?	**(20)**
7.	Problem solving	➢ what problems did you have and how did you solve them?	**(20)**

Total = 100 marks

Student sheet – assessment

	Skill	Team comments	Assessment	
			Team	Teacher
1	planning		/10	/10
2	choosing the correct equipment		/10	/10
3	team work		/10	/10
4	setting up equipment		/10	/10
5	ray diagrams	label your ray diagrams and attach to this sheet	/20	/20
6	describing		/20	/20
7	problem solving		/20	/20
		Total	/100	/100

Student sheet – instructions

- Using the equipment provided set up the first example on your list.
- Draw around the equipment. Include light rays in your diagram and remember to use arrows to show the direction of the light rays.
- When complete show the teacher in charge and describe, using keywords, what is happening.
- Return the equipment and set up your next example.

A. Reflection using a plane mirror.

B. Light rays converging.

C. Reflection along the normal line.

D. Refraction using a transparent block.

E. Refraction using a lens.

F. Light rays diverging.

G. Total internal reflection.

Student sheet – problem solving

Where appropriate draw annotated diagrams to explain your answers.

1 How can mirrors help you to see into hidden corners?

2 How is a prism positioned in a bicycle reflector?

3 Can you demonstrate which type of mirror is best for shop security?

4 Which lens has the shortest focal length?

5 Tom says that water droplets act like tiny lenses. Is he correct? Can you prove it?

6 Describe how you would demonstrate how a periscope works?